

STUDENT LABORATORY MANUAL

Module 4: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing

Companion: Module 4 Student Guide

Toolchain: VS Code + MSYS2 UCRT64 GCC + FreeGLUT

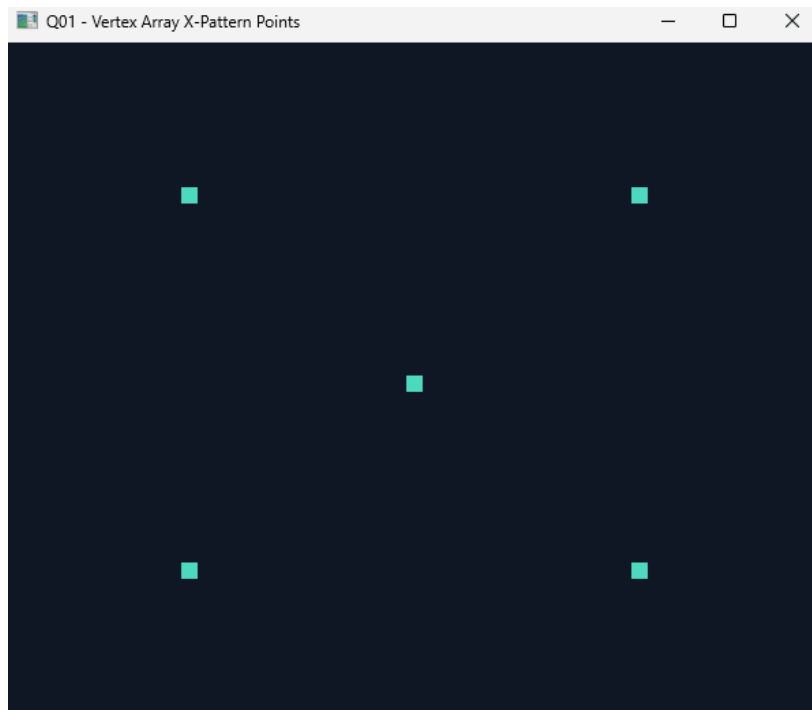
Student Name:	Ebron, Nicole Anne V.
Section:	TN35
Professor:	Dr. Hadji J. Tejuco

GitHub Repository: <https://github.com/nicoleanneebtron-art/Ebron-Nicole-Anne-V>.

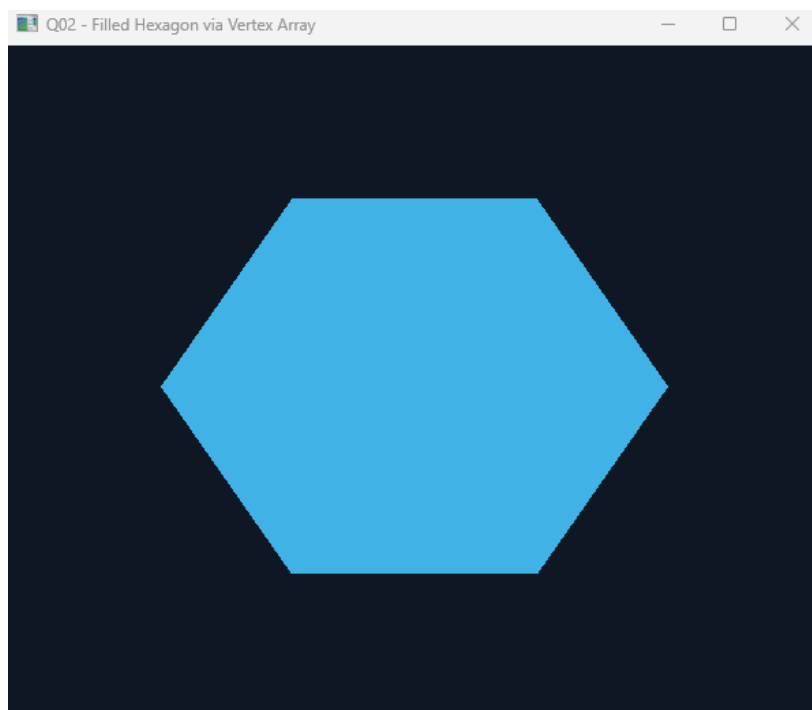
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

Exercise Q01 — Vertex Array X-Pattern Points [Easy]



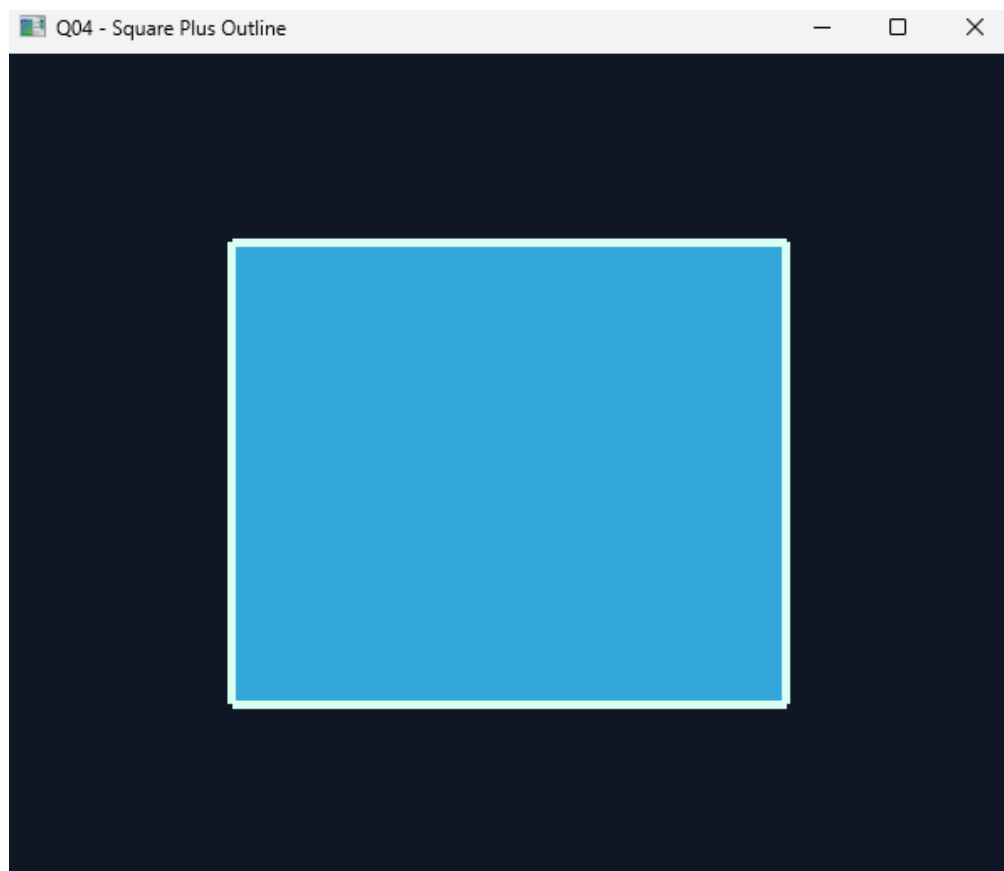
Exercise Q02 — Filled Hexagon via Vertex Array [Easy]



Exercise Q03 — Two Lines, One Array [Easy]



Exercise Q04 — Square Plus Outline, Same Array [Easy]



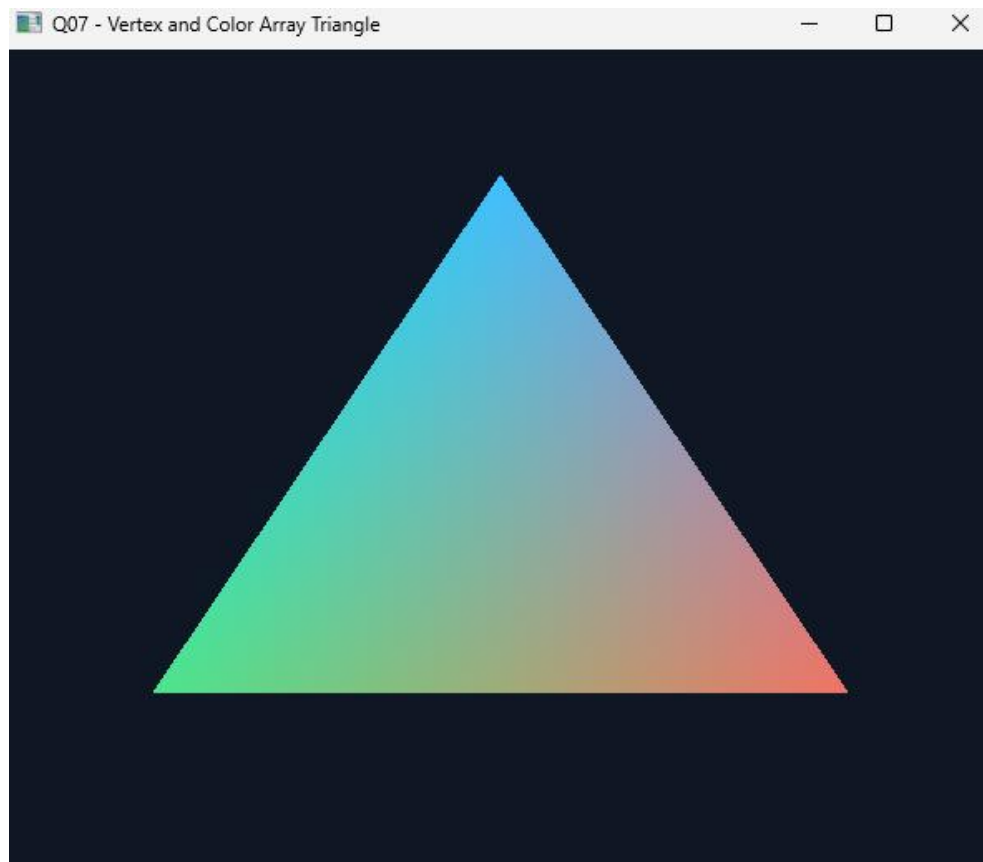
Exercise Q05 — GLint Vertex Data Type [Easy]



Exercise Q06 — glDrawElements Index Order [Easy]



Exercise Q07 — Vertex + Color Array Triangle [Easy]

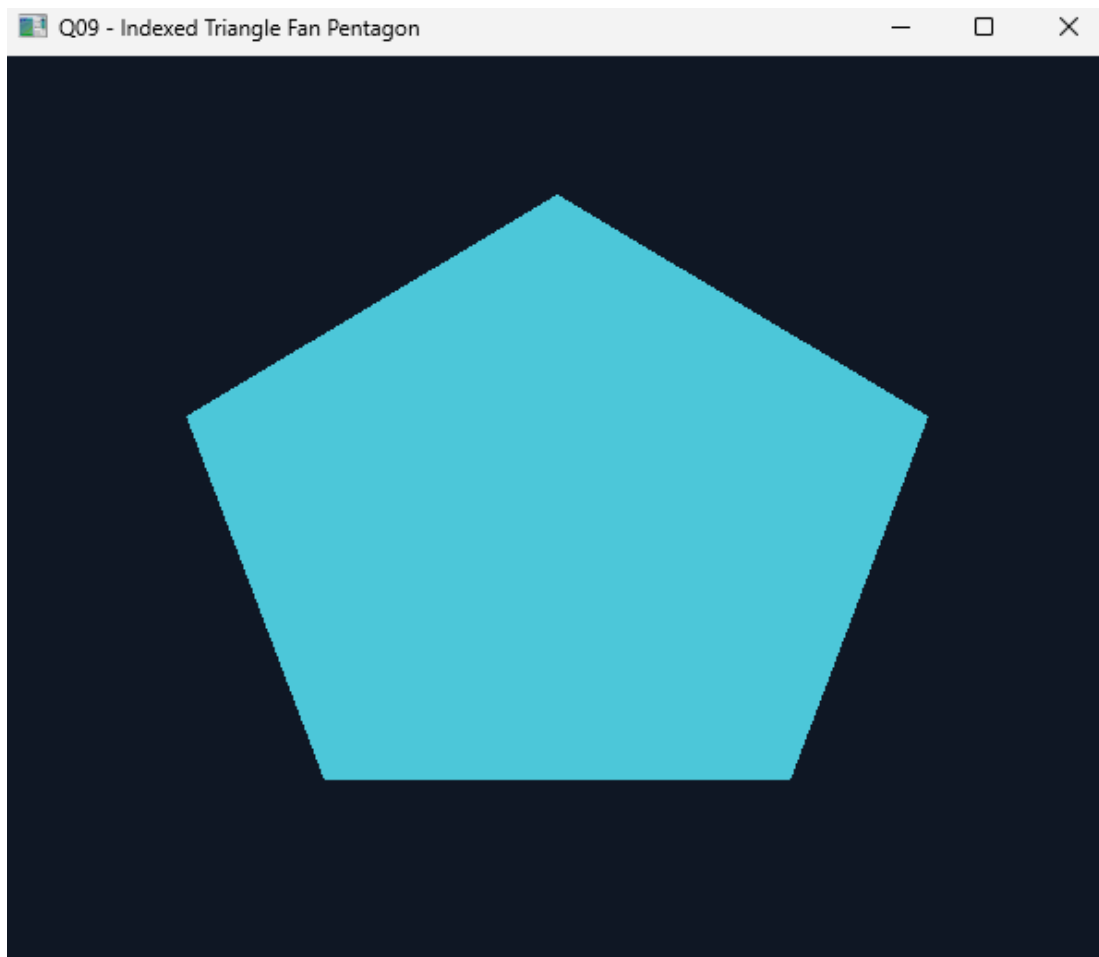


Medium Exercises

Exercise Q08 — Three Triangles, One Array [Medium]



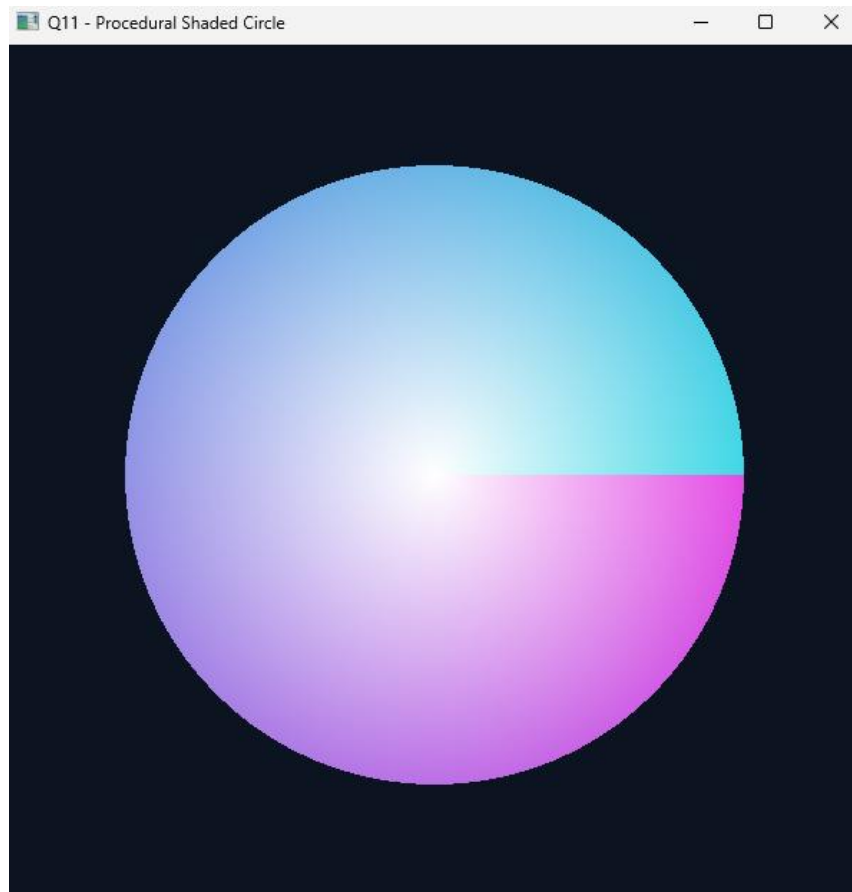
Exercise Q09 — Indexed Triangle Fan Pentagon [Medium]



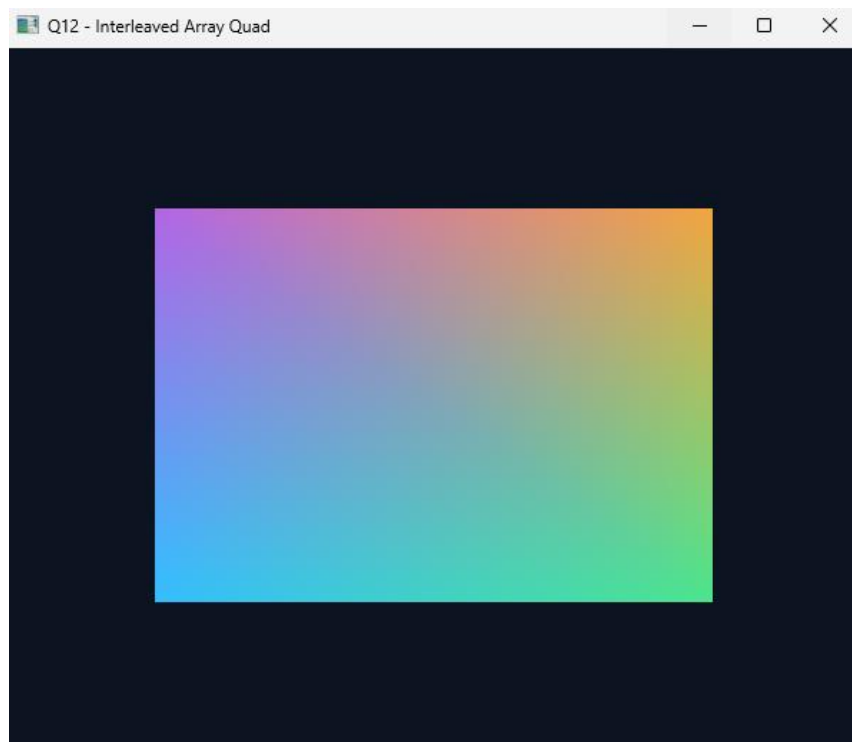
Exercise Q10 — Checkerboard Row via glDrawElements [Medium]



Exercise Q11 — Procedural Shaded Circle [Medium]



Exercise Q12 — Interleaved Array Quad [Medium]



Exercise Q13 — Pinwheel via glDrawElements [Medium]

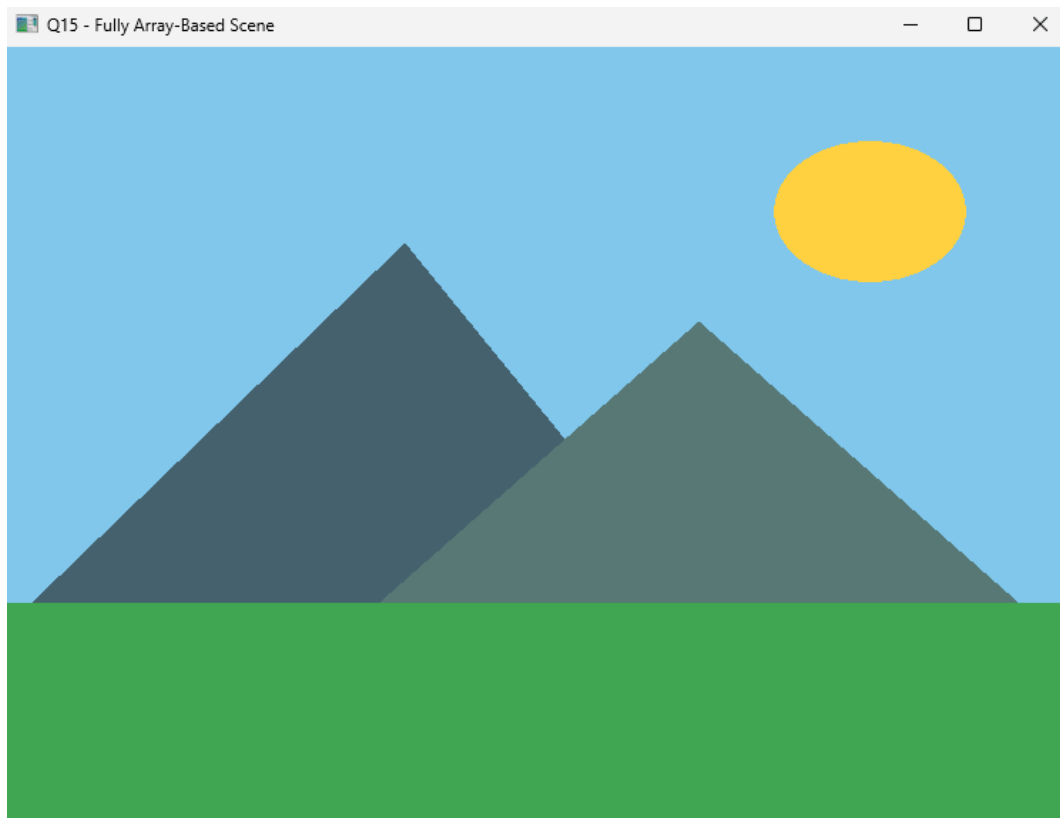


Exercise Q14 — Colored Quad Strip Staircase [Medium]



Hard Exercises

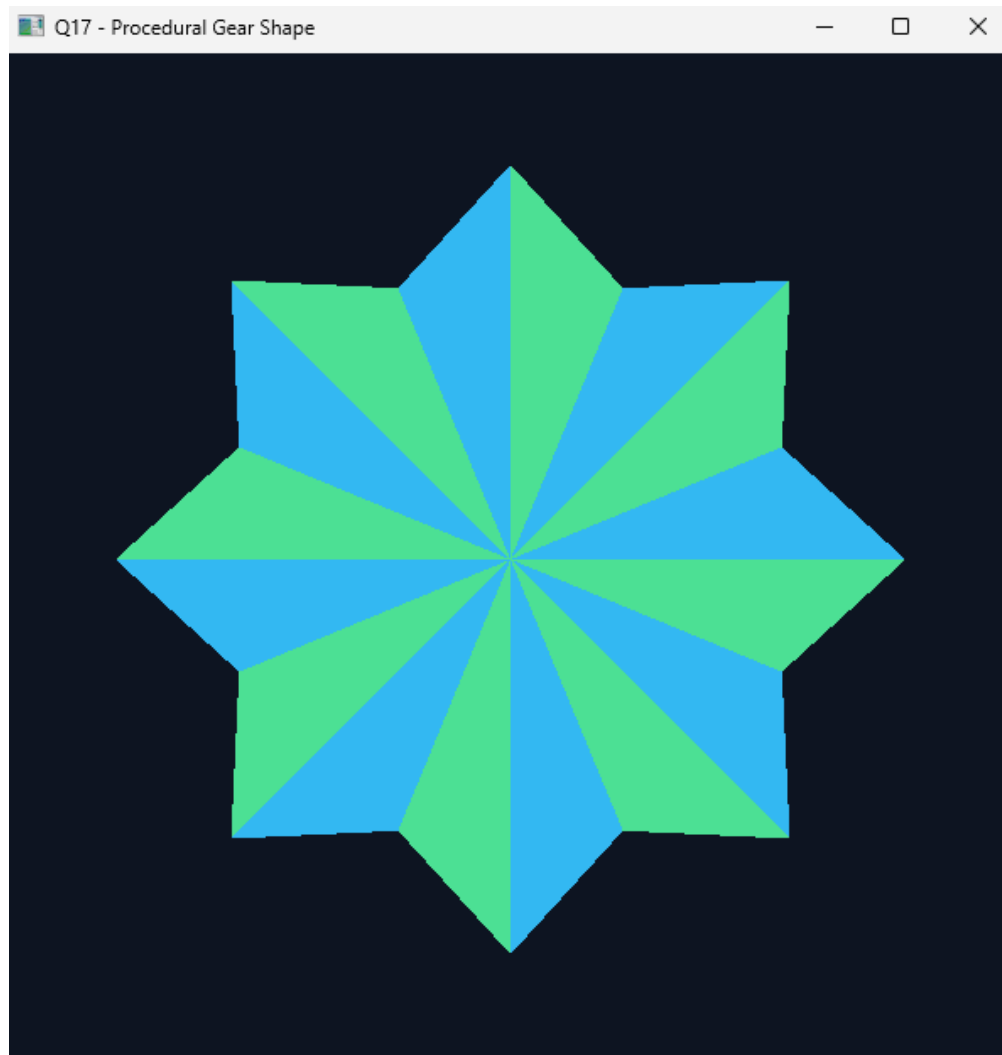
Exercise Q15 — Fully Array-Based Scene [Hard]



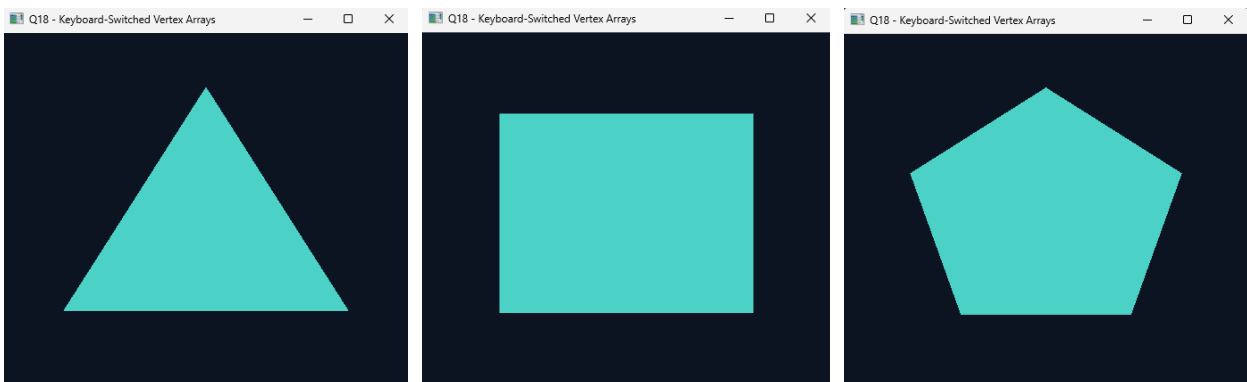
Exercise Q16 — glDrawArrays vs. glDrawElements, Side by Side [Hard]



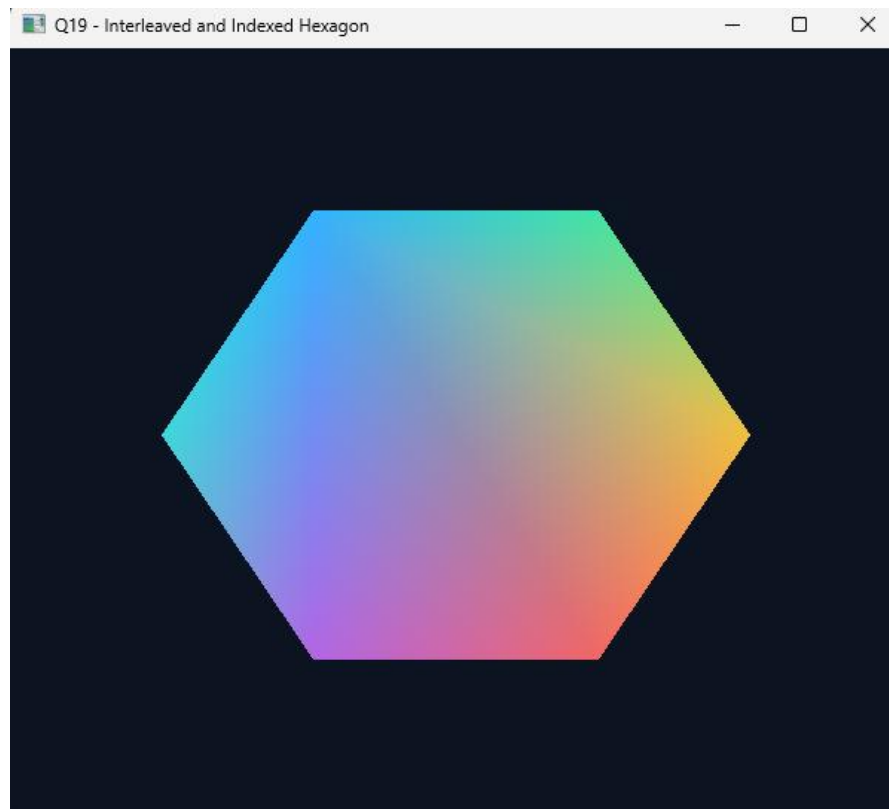
Exercise Q17 — Procedural Gear Shape [Hard]



Exercise Q18 — Keyboard-Switched Vertex Arrays [Hard]



Exercise Q19 — Interleaved + Indexed Hexagon [Hard]



Exercise Q20 — Capstone: Procedural Flower Scene [Hard]

